Application of Green Roof Plants in the CO_2 Sequestration and Day/Night Temperature Reduction: An Experimental Study

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Abstract. Harnessing rooftop greenery for carbon capture and day/night temperature performance is essential to ensure long-term sustainability. Many countries such Indonesia, face serious challenges in environmental management due to high levels of human activity, particularly in the domains of transportation, energy utilization, and infrastructure development. These activities have significant impact on increasing greenhouse gas emissions, especially carbon dioxide. One potential nature-based solution is the application of green roofs. In this study, three experimental models were fabricated and tested according to the climate conditions of Depok city, West Java, Indonesia to achieve temperature reduction and CO_2 sequestration from two green roof modules and a traditional roof module. Each green roof modules utilized *Melampodium Divaricatum (Rich)* and *Noregelia L.B.Sm* as representative C3 and CAM plants, respectively. The study demonstrated that the *M. Divaricatum* module (Model-C) effectively reduced the day/night temperature differences (dT) by 1.5°C during the diurnal period and by 0.4°C during the nocturnal period. In contrast, the traditional roof module (Model-A) showed a maximum increase of 0.4°C during the day and 0.3°C at night. Model-B which used *Noregelia sp.*, exhibited a different pattern from Model-C. It achieved a smaller dT, facilitating limited heat reduction through the soil and substrate layers, resulting in a daily temperature reduction of 0.3°C, however a slight night heat increases of up to 1.2°C. Moreover, the study indicated that *M. Divaricatum* had a lower CO_2 capture capacity of 0.584 $ppm.day^{-1}.leaf^{-1}$, while *Noregelia sp.* exhibited capture rate of 0.667 $ppm.day^{-1}.leaf^{-1}$, highlighting a physiological difference between the two species. Further studies should include continuous data collection period, particularly during night-time, to better assess the full CO_2 sequestration potential of *Noregelia sp*. Additionally, incorporating plants species with C4 photosynthesis pathways could enable comparative analysis of CO_2 capture efficiency and and diurnal temperature fluctuation. Such an approach could enhance roof thermal and contribute to a greater reduction in indoor temperature.

Keywords. rooftop greenery; CO_2 capture; temperature reduction; *Melampodium divaricatum (Rich); Noregelia L.B.Sm.*

INTRODUCTION

Modern green roofs, known as vegetated green roofs [1] or ecological roofs, are new isolated, man-made green areas that consist of membranes and substrates specially designed as planting media and consist of a collection of plants placed on top of buildings or other structures that have innovations. According to research [2], newly created green areas will be separated from intact soil media, creating a natural "niche" and causing strong wind exposure. These areas also receive direct sunlight and experience varying rainfall intensities. Therefore, a good irrigation system is required [3]. Green roof technology was created because of the continuous development of modern building materials and irrigation techniques [4].

Green roofs are considered to be a more popular rainwater management technology than conventional impermeable roofs. Green roofs can retain greater amounts of rainfall, which is eventually returned to the atmosphere through evapotranspiration, and allow it to flow more slowly [5], [6], [7]. According to [8], [9] the function of water infiltration depends on the type of green roof plants, the total volume of the substrate, its composition, and the type of rainfall.

The partial or complete implementation of green roofs can save energy costs for buildings and cities. For instance, a city with sufficient green roofs experienced an overall decrease in ambient temperature in summer [10], [11]. Even the use of green roofs on a building reduces cooling and heating energy requirements in summer and winter because green roofs act as insulators [12], [13], [14] found that green roofs have pore mass, which acts as a noise reduction up to 10 db.

Photosynthesis is a biological process that is very important for all life, providing food and most of the Earth's energy sources [15]. They explained that there are three photosynthetic pathways in rooted plants for assimilating CO_2 gas in the atmosphere, C3, C4, and Crassulacean Acid Metabolism (CAM) [16], [17]. Research from [18] suggested that CAM and C4 photosynthesis evolved as additions to classical C3 photosynthesis around 20–30 million years ago. CAM and C4 photosynthesis are water-efficient by concentrating CO_2 using the dark reaction of photosynthesis. Currently, C4 and CAM plants have filled important ecological niches in grasslands, rainforests, and dry ecosystems.

In particularly, C4 plants dominate grasslands, accounting for nearly 25% of the total primary land area [19], while CAM plants contribute nearly 50% of plants in some dry and semi-dry land areas of the world [20]. C4 plants such as corn (*Zea mays*), sugarcane (*Saccharum officinarum*), and sorghum (*Sorghum bicolor*) account for 22% of the 18% most common food crops grown by humans [21].

The useful of open green spaces on building roofs is significant in maintaining biodiversity, especially in urban areas. For example, hotels with green roofs can attract tourists, resulting in higher room rates for rooms facing the garden terrace. In addition, chefs can easily find herbs and vegetables grown near their hotel restaurants at lower costs and with guaranteed freshness. [22] explains that green roofs offer important esthetics for office workers, reducing fatigue and increasing productivity. Green roofs also filter many carbon pollutants and nutrients needed by plants from the atmosphere, which are then carried away by rain before reaching rivers or lakes [23]. Table 1 explains the green roofs on buildings that function as ecosystems and life cycles.

TABLE 1. Role of green roof on Building (adopted from [24])

Benefits	Green Roof Advantages			Ecosystem sustainability		
	General	Private	Specific	Regular	Culture	Supporter
Rainwater Harvesting	✓	✓		water saving		
Water Infiltration	✓		water	water filter		cycle
UHI	✓			climate		
Energy Building		✓		climate		
Noise reduction		✓		noise	esthetics	
Air quality	✓			air		cycle
Biodiversity	✓			pollination	education	cycle
Fire prevention		✓			esthetics	
Gardens roof		✓	foods		education	soil saving
Education		✓			education	
Carbon Sequestration	✓			air		

Green roof strategies as passive cooling designs have been widely adopted in many contries in recent decades. In particular, weather conditions greatly determine the cooling performance of green roofs [13], field studies are generally limited to evaluating the effects of green roofs with a focus on surface temperatures during specific seasons. Therefore, research is needed to identify specific elements of green roofs that can be used to predict and simulate green roof performance. Research on green roof is dominated by the northern part of America and Europe, which are located in the northern hemisphere, while the southern hemisphere still has minimal research [25]. Therefore, green roof research should be carried out in the context of the region, namely, the tropical climate, especially in the southern hemisphere, such as Indonesia.

Green plants have a transpiration effect on the relatively small changes in room temperature produced by green roofs [26]. Green roof technology continues to advance, one example being the identification of the potential for CO_2 emission capture by extensive plant species with a height range of 6-25 cm and a leaf area index (LAI) range of $0.001 < LAI < 5.0$ [27]. This experiment study is works to new green roof designs, namely (a) selection of plant species for rooftop greenery garden with a plant height ranging from 15 to 30cm (b) selection of plant species with two photosynthesis pathways such as C3 and CAM, which have optimal minimum stomatal resistance (MSR) values in an effort to capture high CO_2 gas, (c) selection of soil media with a substrate layer from biomass in an effort to reduce the day/night temperature on outer surface roof and inside the room. The concept of experimental shows as the models of green roof design and greenery garden plant types is presented in Figure 1.

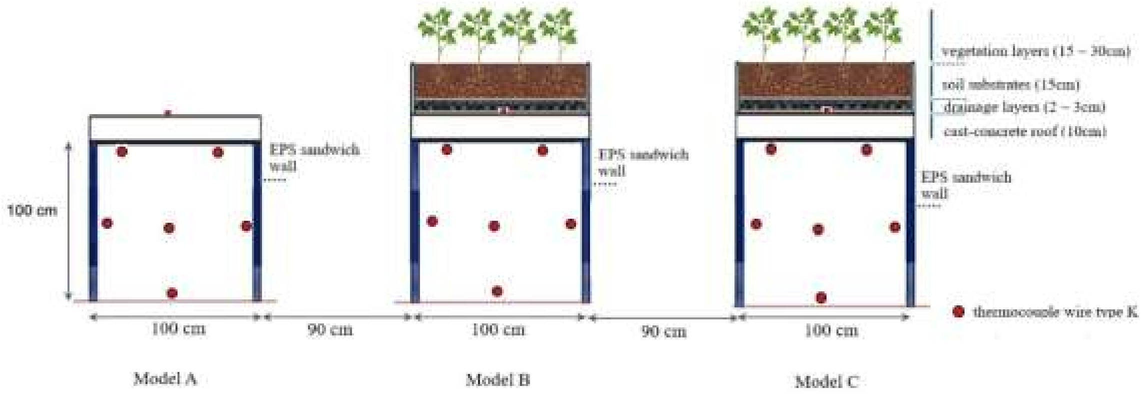


FIGURE 1. Schematic of the experimental models for traditional and green roofs.

Indonesia is an archipelagic country with a population of 278,696,200 [28] and recorded by the Central Statistics Agency with an economic growth of 4.94% in the third quarter of 2023. According to a report from the Ministry of Energy and Mineral Resources, Indonesia has been intensively using fossil resources such as coal, oil, and natural gas for the past 10 years. Human activities that use these resources will have an impact on the amount of energy needed on a national scale and increase GHG emissions [29]. Therefore, developing sustainable knowledge and technology that can mitigate and reduce carbon emissions in the environment is necessary. This is the challenge in achieving the Net Zero Emission target [30].

Based on the literature survey, there is a noticeable lack of research and model experimental studies on the applicability of harnessing rooftop greenery in the performance of CO_2 capture and day/night temperature reduction. This study presents an experimental evaluation of the usage of a locally designed building model to evaluate the extensive green and traditional roofs in Depok city, Indonesia.

Extensive Plant Characterization

Plants were characterized through morphological testing, such plant identification through species testing. Plant determeination was conducted at the Depokensis Herbarium, Biota Collection Room, Department of Biology, Faculty of Mathematics and Natural Sciences, Universitas Indonesia. Part of the characterization and identification of potential local C3 and CAM plant types. Data analysis included leaf area index (LAI), leaf anatomy, leaf reflectivity, minimum stomatal resistance, temperature, humidity and CO_2 binding capacity [31]. Table 2 shows the initial identification results for C3 and CAM plant types, with plant selection based on leaf area size (minimum 3 and maximum 9 cm^2) within height dimensions not exceeding 30 cm. The special characteristics of these plants are that they can survive in extreme conditions on building roof, can survive high temperatures and rainfall in tropical climates such as Indonesia and have evapotranspiration capabilities, and high CO_2 capture rates [32]. In addition, studies [33] related to the morpho-physiology of various potential local plants are needed to determine the most optimal plant modulation in green roof design. Therefore, the photosynthesis process is classified based on the amount of carbon and compounds produced in the CO_2 fixation process, which is divided into three pathways, Table 3.

TABLE 2. Plants characterization

Latin name	Family	Local name	Average plant dimensions (L x W x H cm)	Photosynthesis pathways	Visualization
Melampodium divaricatum (Rich).	*Asteraceae*	*Bunga seribu bintang*	(20 x 20 x 25)	C3 *3-phosphoglycerate*	
Neoregelia L.B.Sm.	*Bromeliaceae*	*Bromelia*	(20 x 20 x 15)	CAM *4-oxaloacetate+ malic acid*	

TABEL 3 Characterization of three types of photosynthetic pathways (adopted from [34])

Comparative aspects	C3 (3-phosphoglycerate)	C4 (4-oxaloacetate)	CAM (4-oxaloacetate + *malic acid*)
Habitat	The climate is temperate and humid, with cool temperatures.	Tropical and warm climate.	Dry and hot climate.
Carbon fixation	The RuBisCO enzyme catalyzes the reaction between CO_2 and RuBP in the first stage of the Calvin cycle to form two three-carbon molecules known as PGA (phosphoglycerate).	Inside the mesophyll cell, the enzyme PEP carboxylase converts CO_2 and PEP, into a four-carbon known as malate through a series of reactions.	Inside the mesophyll cell, the enzyme PEP carboxylase converts CO and PEP, into a four-carbon compound known as malate through a series of reactions.
Leaf stomatal (occurring carbon fixation)	Open during sunlight and dry conditions.	Open slightly when the conditions are dry and the sun is hot.	Open at night.
Benefits	• The most efficient photosynthesis occurs under cool, humid conditions. • High CO_2 concentration.	• The most efficient photosynthesis occurs under hot, dry conditions. • The lowest CO_2 concentration. • Lower photorespiration.	• Survival in extremely dry conditions. • Leaf transpiration is relatively slow.
Weakness	• Highly photorespiration. • The photosynthetic efficiency declines.	• Requires more adenosine triphosphate (ATP) energy to produce glucose.	• Plant growth is slower due to lower energy efficiency for conversion into acid and re-storage as CO_2.

They described [6] that extensive green roofs offer the advantages of economical initial investment costs, lower maintenance costs compared to semi-intensive and intensive roofs, and lower watering requirements compared to intensive green roofs. In general, extensive green roofs require a lighter structure and do not require roof or building structure reinforcement. In addition, extensive roofs can be applied to steeper slopes and are technically simple to construct. However, extensive green roofs have relatively limited energy efficiency and rainwater management capabilities [35]. Extensive roofs are the most commonly applied because they require a lightweight structure and simple drainage systems, as well as lower initial investment and maintenance costs [36].

Location and Condition of Experiments

The simulation modeling was selected using open-source software, likely Andremarsh and the location on the roof top at the Manufacturing Research Center (MRC) building, Faculty of Engineering, Universitas Indonesia, because the simulation results enabled the optimization of the performance of the vertical green roof system in response to climate and microclimate changes [37]. This study also conducted a shadow simulation to determine the distance between samples so that shadows falling on the research samples do not affect each other's temperature. A distance of 90 cm was set so that no shadows fell on other roofs, causing deviations during measurement. Figure 2 illustrates the simulation stage conducted on the AndrewMarsh.com shadow simulation application on March 12, 2025. The Davis Instrument weather data are collect for four days on June 11-14, 2025. The climate conditions and solar radiation are shown in Figure 3.

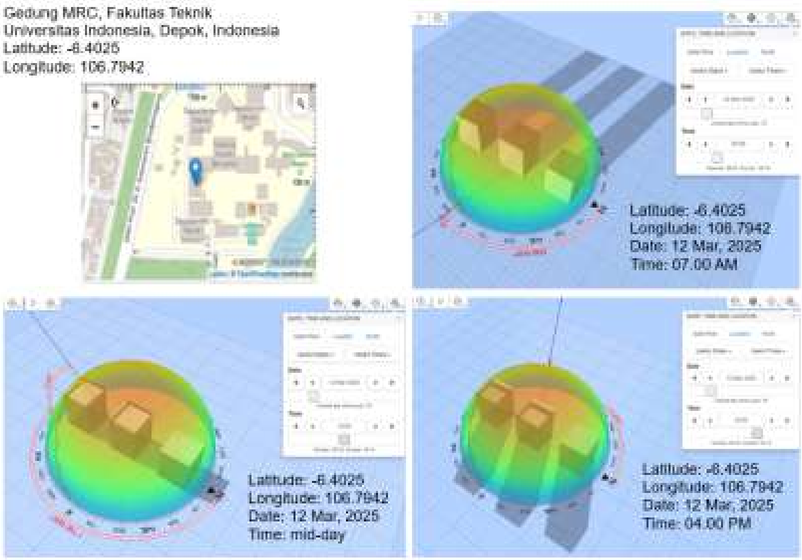


FIGURE 2. Andremarsh simulation
(simulated to the Indonesian west time on March 12, 2025 at 7AM: 12AM: 4PM).

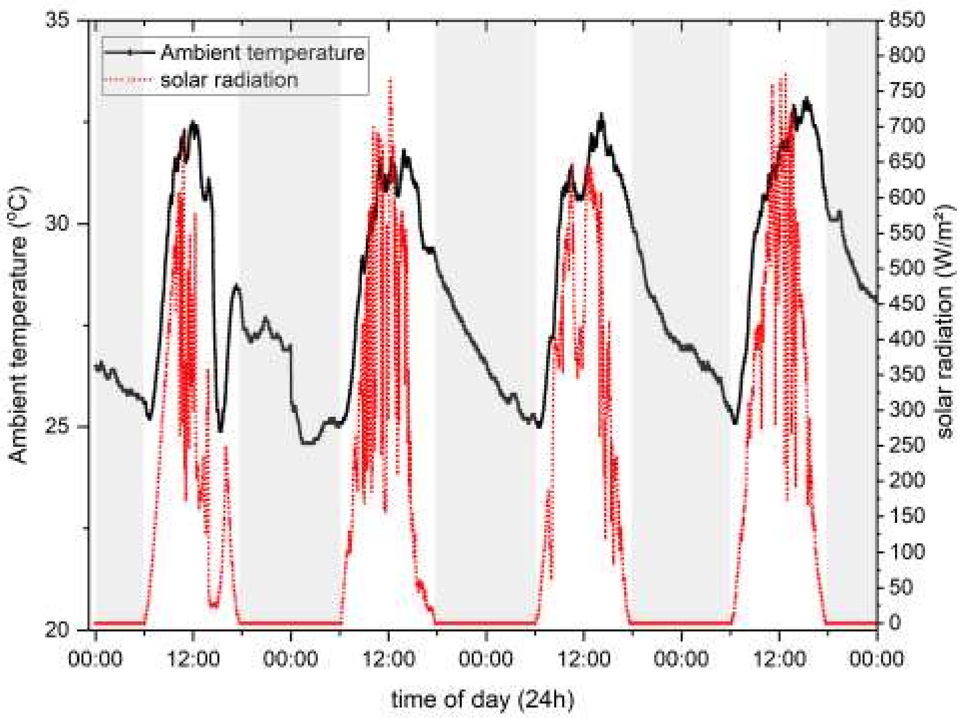


FIGURE 3. Hourly aeration of solar radiation vs local time in Depok city

Thermal load calculations for the traditional roof from module-A and extensive green roofs from modules B and C were performed by determining building components related to green plants, weather sensors using Davis Instrument weather stations, and microclimate sensors including thermocouples, air humidity (RH%), heat intensity (heat flux). Building were performed applying Fourier's Law on heat conductors from equation (1), Q as thermal load (Watts) and equation (2) Fourier on heat insulators as the heat resistance (K/Watts).

$$Q = \frac{k.A.\Delta T}{\Delta x} \tag{1}$$

$$R_{th} = \frac{\Delta T}{Q} \tag{2}$$

Experimental Configuration and Setup

The application of an extensive green roof is to address the research objectives, three measurement variables are used (a) control variables, including concrete roof material layers and weather conditions in a tropical climate, (b) independent variables, including green roof layers: planting media thickness, and two types of extensive green plants (Table 4); and (c) dependent variables are plants performance in high CO_2 capture and thermal performance generated in research data, including outer surface roof temperature, inside surface roof temperature, and room temperature.

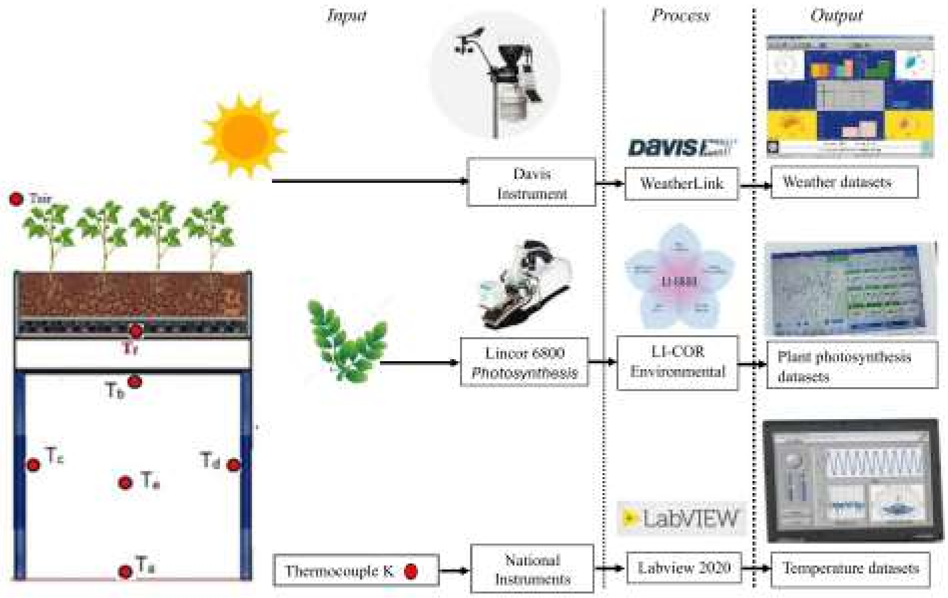


FIGURE 4. Schematic of the experimental setup of the extensive green roof.

This study used the National Instruments as temperature measurements of the effect of independent variables on the surface temperature of the research sample roofing material. The measurement instruments required were temperature sensors and radiation sensors. Figure 4 shows an illustration of the instrumentation used in this study. K-type thermocouple wires were placed at seven points on the green roof module sample and six points on the traditional roof model, with each point representing a specific temperature: (Tair) represents the ambient temperature, (Tf) represents the surface temperature between the planting medium and the upper surface of the concrete roof, (Tb) is the lower surface temperature of the concrete roof, (Tc) and (Td) measure the surface temperature of the walls on the east and west sides, (Te) shows the indoor temperature, and (Ta) is the surface temperature of the floor inside room.

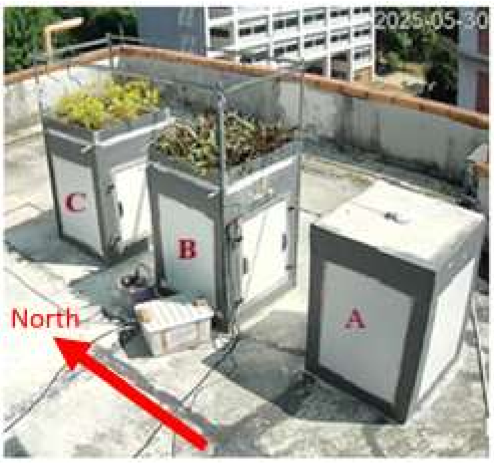


FIGURE 5. Visualization of the experiment models; traditional roof (A), green roof used *Neoregelia L.B.Sm* (B) and green roof with *M. Divaricatum* (C)

This experimental study was constructed for three building models, namely A, B, and C. Table 4 show the model configuration with a total roof volume of 0.1 m^3 for each module and distance of 90 cm between modules to prevent shadows from falling on the roof. Figure 5 and table 5 show the constructing process of the the building structure, walls, and roof. Other experimental studies also applied similar sizes, with each roof module having a volume of 0.081 m^3 [38] and 0.054 m^3 [39].

TABLE 4. Models Constructions

Components	Models Experimental		
	Model-A	Model-B	Model-C
Material roofs	Cast concrete, 100 mm thickness	Cast concrete, 100 mm thickness	Cast concrete, 100 mm thickness
Material walls	Expanded Polystyrene (EPS) sandwich (75 mm)	EPS sandwich (75 mm)	EPS sandwich (75 mm)
Material tiers	Cast concrete	Cast concrete	Cast concrete
Dimensions model (L x W x H cm)	100 x 100 x 100	100 x 100 x 100	100 x 100 x 100
Dimensions roof (L x W x H cm)	100 x 100 x 10	100 x 100 x 10	100 x 100 x 10
Plants	Not available	Melampodium divaricatum (Rich).	Neoregelia L.B.Sm.
substrates thickness (cm)	Not available	15	15

TABLE 5. Roof and substrate parameters

Parameter	Value	Ref
Conductivity coefficient of concrete K225 (kb)	1.6 W/m.K	Indonesia standard No. 03-6389-2000
Dry soil conductivity coefficient (k_{tk})	0.39 W/m.K	[40]
Wood powder conductivity coefficient (k_{tk})	0.932 W/m.K	[41]
Concrete thickness (B_1)	0.10 m	Experimental design
Soil thickness (B_2)	0.15 m	Experimental design
substrate thickness (dx)	0.05 m	Experimental design
Roof volume (V)	0.10 m^3	Experimental design

Method of Collecting Data

This method was used to obtain data on surface temperature differences in green roof material elements and to obtain data on CO_2 capture performance from plants through photosynthesis in plant leaves. As illustrated, the measurements were taken on concrete extensive green roof to produce primer data. The data collection stage consists of three steps: the input process, which includes determining the research subject and research variables, the process stage, which includes research instruments; and the output stage, which is the data generated from the measurement process. Figure 6 shows the influence between the research and the database to obtain the data output such as primer and comparative data.

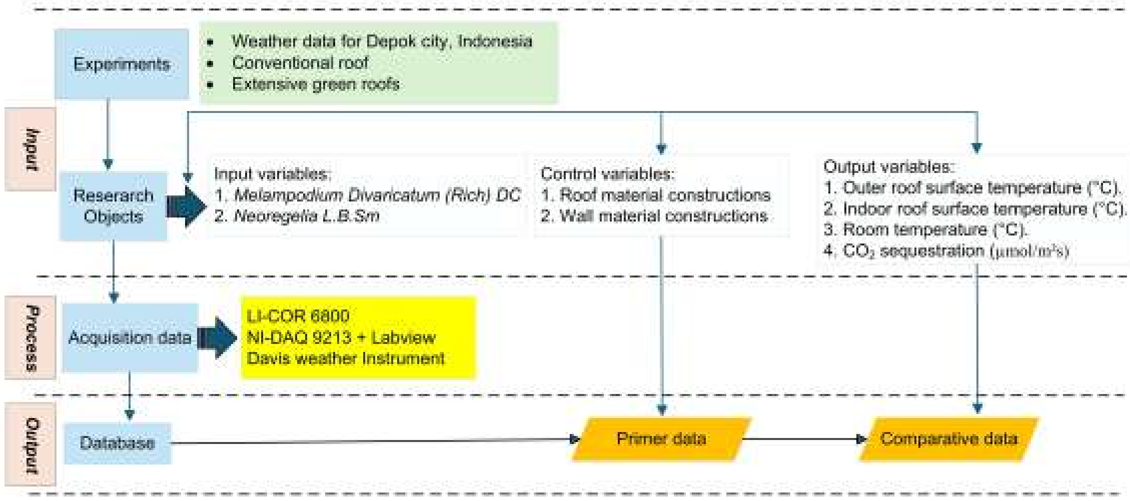


FIGURE 6. Flowchart of the experimental measurements

The leaf gas exchange data obtained using the LI-COR 6800 photosynthesis system were utilized as an initial basis for estimating the overall CO_2 sequestration capacity of the plants during photosynthesis. In this study, measurements were conducted on two types of plant species representing photosynthesis pathways: *Melampodium Divaricatum (Rich)*, commonly known as *Bunga seribu bintang* from the *Asteraceae* family and classified as a C3 plant, and *Neoregelia L.B.Sm*, known locally as *Bromelia* from the *Bromeliaceae* family and categorized as a CAM plant. Both species were monitored over a four-day period, from June 11 to 14, 2025, with four measurement sessions each day: early morning (4:00 - 5:00 AM), morning (8:00 - 10:00 AM), slightly mid-day (11:00 AM - 1:00 PM, and afternoon (4:00 - 5:00 PM). Stomatal resistance (Rc) serves as a key parameter in determining the degree of resistance to gas exchange-specifically CO_2 uptake and water vapor release through the leaf stomata. The following equations were employed to convert stomatal conductance data into gas pressure ad absolute temperature values [26]:

$$R_c = \frac{1}{gsw} \times \frac{P}{R \cdot T} \quad (3)$$

The value is obtained from stomatal conductance (gsw) measured using a LI-COR 6800 device, with units of mol/m²s. The stomatal resistance (Rc) values were calculated from the conversion of stomatal conductance (gsw) based on standard plant physiology equations.

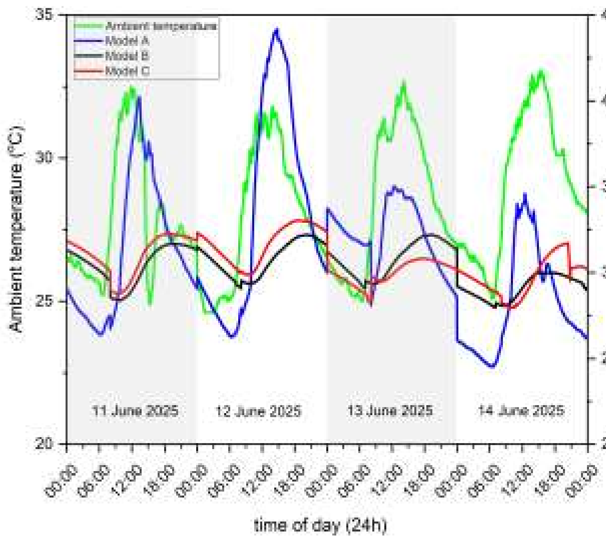
RESULTS AND DISCUSSION

Roof Day/Night Temperature Performance

The temperature generated at point (Tf) on the building roof module shows the total heat absorption capacity (heat flux) from solar radiation. The traditional roof receives solar radiation directly without any barriers. Concrete, a conductor, absorbs heat and stores it until it reaches a certain temperature within a certain time. Thus, the roof reaches its peak temperature at the same time as the outside temperature because concrete has high conductivity to absorb heat quickly, providing insight into the characterization of solar radiation in relation to the ability of roof materials to transfer heat. The (Tf) temperature on model-B is slightly lower than that on the model-C. The soil layers on both green roofs have the same composition and thickness, so the effect of the temperature difference is also similar. The composition and thickness of the soil layer on both green roof modules (GRMs) cause a temperature difference, whereby the plant layer acts as heat insulation. The results of this experiment provide an understanding of how plants dynamically influence thermal changes due to weather factors by analysing the factors that contribute to temperature variation.

Figure 7 shows the effect of green roofs on room temperature (Te), which gives a maximum room temperature difference between green roofs and traditional roof of 4.15ºC. This occurred on June 12, 2025, at the peak ambient temperature (Tair) between the 720th and 750th minutes, or at 12 noon. At night, the graph data show that there is smaller temperature difference between green and traditional roofs. However, a significant gradient difference of 3.45ºC on June 14, 2025, at 1400 minutes or 11.30 PM due to an increase in the ambient temperature of up to 2ºC compared to previous days.

Table 6 shows a comparison of day/night temperature performance based on the room and ambient temperatures. The effective dT was achieved to 14 June, 2025, by the model-C with *M. Divaricatum* its successful as passive cooling capability, reducing the room temperature by (-) 1.5ºC at the day and (-) 0.4ºC in the night compared to the ambient temperature, while the traditional roof produced the smallest temperature reduction in the layer under the roof as the room temperature, which was (+) 0.4ºC at the day and (+) 0.3ºC at the night. However, the model-B, nearly succeeded in providing passive cooling for (-) 0.3ºC at the day and (+) 1.2ºC in the night**.** These results of the experimental study almost obtained the effectiveness of temperature performance compared with the results studies by [42], which obtained dT values from (+) 1.2 at the day and (+) 0.8°C in the night, and [43] day/night temperature by (-) 0.6 at day and (-) 0.9°C at night.



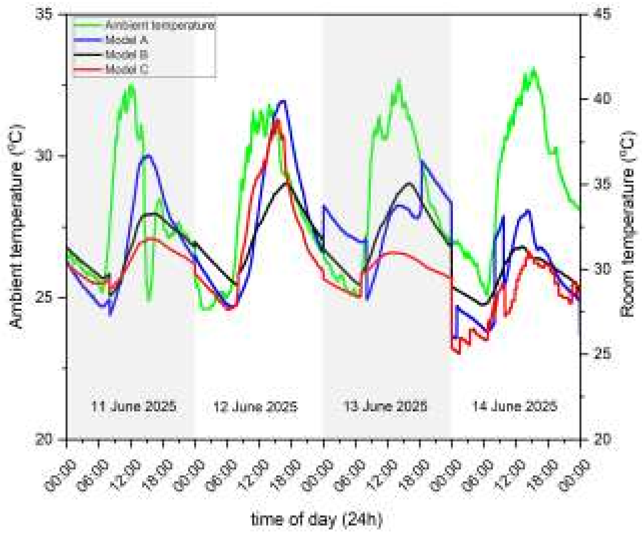


TABLE 6. Day/Night temperature performance

Effectiveness roofs		11 June Day Mean	SD	11 June Night Mean	SD	12 June Day Mean	SD	12 June Night Mean	SD
T_e (ºC)	Model A	30.2	3.6	30.7	1.8	33.7	4.6	31.5	2.8
	Model B	31.2	1.8	31.3	1.0	32.3	2.1	31.9	1.5
	Model C	30.5	0.7	30.3	0.7	34.1	3.6	30.2	1.7
$T_{ambient}$ (ºC)		29.0	2.5	26.7	0.6	29.6	1.9	26.3	1.4
dT (ºC)	Model A	1.2	1.1	4	1.2	4.1	2.7	5.2	1.4
	Model B	2.2	-0.7	4.6	0.4	2.7	0.2	5.6	0.1
	Model C	1.5	-1.8	3.6	0.1	4.5	1.7	3.9	0.3

Effectiveness roofs		13 June Day Mean	SD	13 June Night Mean	SD	14 June Day Mean	SD	14 June Night Mean	SD
T_e (ºC)	Model A	32.1	1.6	33.8	1.4	30.9	2.3	28.1	1.3
	Model B	32.8	2.0	31.4	1.4	30.2	1.1	29.0	0.7
	Model C	30.4	0.7	29.4	0.6	29.0	1.5	27.4	1.6
$T_{ambient}$ (ºC)		30.1	2.2	26.9	1.4	30.5	2.4	27.8	1.5
dT (ºC)	Model A	2.0	-0.6	6.9	0	0.4	-0.1	0.3	-0.2
	Model B	2.7	-0.2	4.5	0	-0.3	-1.3	1.2	-0.8
	Model C	0.3	-1.5	2.5	-0.8	-1.5	-0.9	-0.4	0.1

Thermal Effectiveness of Soil Layers

The effectiveness of the soil layers of the three roof types can be evaluated using equation (4), where the q value is obtained using Equation (1).

$$\eta = \frac{q_{concret\ roof} - q_{soil\ layers}}{q_{concret\ roof}} \quad (4)$$

The principle of energy balance must also be applied to evaluate heat in soil layers and plants. In the soil layer, the qt value was determined by measuring dT and dx in the soil medium, and the qb value was determined by measuring dT and dx on the upper-lower surface concrete roof. Table 5 presents the soil specifications in this study. The evaluation of thermal effectiveness in three roof modules is presented in Table 7, which shows that model-C has a fairly better insulating ability compared to the model-B. Hence, the thermal quantity (q) value in green roof applications with a positive (+) possessed by the planting medium means that the soil medium has sufficient water content, while the negative (-) produced by the concrete roof means that the material releases heat to reduce the roof temperature.

TABLE 7. Roof thermal evaluation of the experimental study

Effectiveness thermal		11 June		12 June		13 June		14 June	
		q_t	q_b	q_t	q_b	q_t	q_b	q_t	q_b
q (Watt/m^2)	Model B	37.9	16	10.9	20.8	9.4	22.4	1.3	14.4
	Model C	13.8	-3.2	10.9	-6.4	3.6	-1.6	-2.3	-11.2
	Model A		4.8		12.8		33.6		4.8
η	Model B	-1.3		0.47		0.58		0.91	
	Model C	-3.31		-0.71		-1.25		-0.79	

Leaf Performance for the CO_2 Sequestration

CO_2 capture measurements were performed by recording the net photosynthesis rate (µmol.CO_2/m^{2}s) from equations (5,6) and the stomatal conductance (gsw) was measured using the LI-COR 6800 instrument. According to a previous study [44], the stomatal conductance of leaves is directly proportional to the square root of the H_2O evaporation process in the leaves. The carbon fixation process from the Calvin cycle through RuBisCo cells (ribulose-1,5-bisphosphate carboxylase-oxygenase) is responsible for binding carbon into glucose [45] with higher CO_2 gas concentrations. The CO_2 emission rate value was calculated using equation (5), $\frac{dC}{dt}$ (mol/m^3.s).

$$gsw = \frac{PV}{RTA} x \frac{dC}{dt} \tag{5}$$

$\frac{dC}{dt}$ (mol/m^3/s) of carbon dioxide levels will be measured as part of the CO_2 assimilation process:

$$\text{assimilation } CO_2 = \frac{Flow\ x\ (CO_2 sample - CO_2\ referensi)}{(\frac{dC}{dt})} \tag{6}$$

Table 8 and 9 present the results of gsw, radiation intensity, air temperature, leaf temperature, relative humidity, and Rc calculations for the two plants. These values were investigated to understand the physiological patterns of leaf stomata in response to environmental variables in each plant type.

TABLE 8. Leaf stomatal conductance (gsw) against leaf stomatal resistance (Rc) for *M. Divaricatum*

date	Time day session	gsw (mol/m²s)	Solar radiation (W/m²)	$T_{ambient}$ (°C)	T_{leaf} (°C)	RH% (°C)	R_c s/m
June 11, 2025	Early morning	0.301	0.0	26.8	26.6	90	135.1
	Morning	0.172	310.5	33.5	32.7	80.6	231.1
	Noon	0.140	633.1	34.6	33.8	71.2	283.0
	Afternoon	0.134	225.3	29.9	29.5	81.8	300.3
June 14, 2025	Early morning	0.127	0.0	26.1	25.4	99	320.8
	Morning	0.133	411.1	34.2	33.4	88.7	298.3
	Noon	0.147	676.9	34.8	33.9	73.9	269.4
	Afternoon	0.099	126.0	27.0	26.5	81	410.3

Based on the results of calculating stomatal resistance values using Equation (3), the performance of both types of plants can be understood. The stomatal resistance value of *M. Divaricatum* plant shows an average stomatal resistance value of 275.4 s/m, whereas that of *Neoregelia,sp* plant showed a higher average of 2,059.9 s/m. The lower Rc to indicates that the stomata of these plants are more often fully open, thereby accelerating the diffusion of H_2O and CO_2. This is in accordance with the physiological characteristics of *M. Divaricatum* that are active during the day, namely, having the best photosynthetic efficiency occurring in cool, humid conditions and high CO_2 uptake concentrations.

On the other hand, *Neoregelia,sp* plants showed very high Rc values, even reaching values above 3,000 s/m in the afternoon, and minimum values in the range of 754.8 s/m. The large range of variation indicates that environmental conditions greatly influence the reactivity of leaf stomata. Rc values are very high from morning to afternoon because the photosynthesis pathway opens its stomata during the dark phase, i.e., at night. This is different from the daytime, when resistance increases as the stomata tend to close to avoid water loss. The growth of *Neoregelia,sp* plants is slower due to lower energy efficiency during the light phase for conversion into acid and re-storage as CO_2.

Significant difference in Rc values in physiological characteristics between the two plants. *M. Divaricatum* plant consistently show lower Rc value, especially during the light phase when the stomata are fully open and the photosynthetic efficiency is optimal. Conversely, *Neoregelia,sp* show higher and fluctuating Rc values, indicating that the plants do not perform photosynthesis to avoid excessive transpiration.

TABLE 9. Leaf stomatal conductance (gsw) against leaf stomatal resistance (Rc) for *Neoregelia, sp*

date	Time day session	gsw (mol/m²s)	Solar radiation (W/m²)	$T_{ambient}$ (°C)	T_{leaf} (°C)	RH% (°C)	R_c s/m
June 12, 2025	Early morning	0.020	0.0	26.1	26.2	96	2,037.3
	Morning	0.031	117.2	34.1	34.2	93.7	1,280.2
	Noon	0.017	569.9	34.8	34.6	84.1	2,329.1
	Afternoon	0.010	108.7	32.8	32.7	80.7	3,993.2
June 13, 2025	Early morning	0.054	0.0	26.0	26.1	83.5	754.8
	Morning	0.019	442.6	32.0	32.0	78.4	2,103.1
	Noon	0.019	599.6	33.9	33.9	73.4	2,090.1
	Afternoon	0.021	76.9	33.8	33.9	72.5	1,891.6

Table 10 and 11 present the results of calculations of plant performance in photosynthesis by capturing CO_2 gas and transpiration H_2O, as well as the effectiveness of H_2O use from the soil, such as Water Use Efficiency (WUE). These values were analyzed to understand the photosynthetic of both types of plant types.

TABEL 10 Diurnal Variation in Photosynthesis Parameters (***CO_2*** Sequestration, Transpiratioan and WUE) of *M. Divaricatum* Leaves Measured by Li-COR 6800

date	Time	CO_2 Sequestration µmol/m²s		Transpiration H_2O (mol/m²s) x10^{-3}		WUE µmol/mol x10^{-4}	
	day session	Mean	SD	Mean	SD	Mean	SD
June 11, 2025	Early morning	-0.047	0.09	0.97	0.18	-0.005	0.010
	Morning	0.014	0.08	3.05	0.30	0.001	0.003
	Noon	0.032	0.06	3.07	0.41	0.001	0.002
	Afternoon	0.031	0.14	1.43	0.14	0.002	0.011
June 14, 2025	Early morning	-0.016	0.05	2.25	0.82	-0.020	0.140
	Morning	0.011	0.04	3.97	0.75	0.001	0.001
	Noon	0.044	0.06	2.50	0.40	0.002	0.003
	Afternoon	0.035	0.08	1.84	0.79	0.002	0.003

Based on the results, *M. Divaricatum* showed an average daily photosynthesis rate of 0.027 CO_2 µmol/m²s on 11 June and 0.031 CO_2 µmol/m²s for the 14 June, with a total CO_2 absorption of 0.584 ppm.day^{-1}.leaf^{-1}. In contrast, *Neoregelia sp* recorded an average of 0.03 µmol/m²s, with a total daily absorption of 0.667 ppm.day^{-1}.leaf^{-1}. The highest value for *M. Divaricatum* plant was recorded in the afternoon on June 13, at 0.044 CO_2 µmol/m²s, while for *Neoregelia sp* plants, the highest value only reached 0.070 CO_2 µmol/m²s in the morning session on the day.

TABEL 11 Diurnal Variation in Photosynthesis Parameters (***CO_2*** Sequestration, Transpiratioan and WUE) of *Neoregelia, sp* Leaves Measured by Li-COR 6800

Time		CO_2 Sequestration µmol/m²s		Transpiration H_2O (mol/m²s) x10^{-3}		WUE µmol/mol x10^{-4}	
date	Day session	Mean	SD	Mean	SD	Mean	SD
June 12, 2025	Early morning	0.001	0.09	0.02	0.09	0.240	1.38
	Morning	0.070	0.05	0.80	0.20	0.010	0.01
	Noon	-0.040	0.04	0.50	0.20	-0.010	0.01
	Afternoon	-0.030	0.04	0.21	0.06	-0.010	0.02
June 13, 2025	Early morning	0.010	0.08	0.01	0.13	0.460	2.54
	Morning	0.070	0.13	0.38	0.17	0.005	0.14
	Noon	-0.020	0.04	0.56	0.14	-0.010	0.01
	Afternoon	-0.030	0.03	0.77	0.07	-0.003	0.004

Based on the environmental parameters, at fundamental correlation was observed between solar radiation intensity and the photosynthesis rate of *M. Divaricatum*. Higher radiation values, particularly during the noon session (676.9 W/m² on June 14), were associated with an elevated CO_2 assimilation rate of 0.044 µmol.m^{-2}s^{-1}. Conversely, in the early morning session with radiation of 0.0 W/m², the photosynthesis rate tended to be negative or very low. This study shows that C3 plants, such *M. Divaricatum* have high leaf stomata performance, which is usually dependent on light intensity as the main parameter of plant photosynthesis. On the other hand, *Neoregelia,sp* showed a different pattern. Although, the solar radiation values increased during the day, the average CO_2 rate remained low and even negative. This is consistent with the characteristics of CAM metabolism, likely *Neoregelia,sp* which is more active in the night to the early morning. As the results, on June 13 for noon of day session with the solar radiation is to 599.6 W/m², give to the CO_2 rate was negative (-0.020 µmol.m^{-2}s^{-1}) its brief indicating that even though the light from sun was quite high, the plants were not actively to CO_2 sequestration because of leaf stomatal is open in the night or at the day with lowest solar radiation.

CONCLUSION

A preliminary characterization of greenery garden plants for extensive green roof modules with high CO_2 sequestration capacity was conducted. This experimental study evaluated the thermal performance and CO_2 capture capacity of two commonly used ornamental plants in Indonesia, *Melampodium Divaricatum (Rich),* representing the C3 photosynthesis pathways and *Noregelia L.B.Sm* representing in the CAM pathway. Both species were applied in extensive green roofs models to assess their efficiency in moderating roof temperature and capturing atmospheric CO_2. The thermal analysis demonstrated that *M. Divaricatum* green toof module (Model-C) effectively reduced the temperature difference (dT) between the outer and inner roof layers, or (dT), as the thermal performance of the roof module. Based on this study, model-C within the C3 green roof module reduced by 1.5°C during the daytime and 0.4°C at night. In contrast, the traditional roof (Model-A) showed a temperature increase of 0.4°C during the day and 0.3°C at night. The *Noregelia,sp* module (Model-B) exhibited a smaller reduction in Dt, with a daytime decrease of only 0.3°C and slight nocturnal increase of up to 1.2°C. Among the three roof modles, Model-C showed the highest thermal insulation efficiency, with a thermal roof effectiveness ranging from 0.71 to 3.71. This indicates that vegetation-based roofs can significantly reduce heat transfer compared to conventional roofs. Measurements using the Li-COR 6800 photosynthesis system futher revealed differences in CO_2 sequestration between the two plants types *M. Divaricatum* exhibited a CO_2 gas capture rate of 0.584 ppm.day^{-1}.leaf^{-1}, while *Noregelia L.B.Sm* achieved a higher rate of 0.667 ppm.day^{-1}.leaf^{-1}. These results highlight physiological distinctions between C3 and CAM plants in carbon capture efficiency. Overall, the performance variations between the two species provide essential insights for selecting suitable plants for extensive green roof applications aimed at improving thermal regulation and enhancing CO_2 sequestration. Further research should include continuous day-night monitoring to capture the CO_2 absorption potential of CAM plants such as *Noregelia L.b.Sm*. Furthermore, integrating C4 plant species could enable comparative evaluation of photosynthesis efficiency and diurnal temperature control, thereby enhancing the overall thermal performance and sustainability of green roof systems.

ACKNOWLEDGEMENTS

The authors would like to sincerely thank all individuals who contributed to this research. We also gratefully acknowledge the financial support provided by the Universitas Indonesia with Q1 PUTI 2025 with granted number PKS-201/UN2.RST/HKP.05.00/2025. The authors also gratefully acknowledge the Laboratory of PAUPT-NZI, FMIPA Universitas Indonesia, for providing the LI-COR 6800 photosynthesis instrument and technical support during this research. Their support was crucial for the successful completion of this work.

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